BGI Sample Testing Report

1. Project Information

Report No.: TPLa23081106

Project Name	TAMUCC_Frauke Seemann_21Japanese medaka_totalRNA_ RNAseq_PE150_20M_Nobi_AWS_073123			**Project No.**	F23A480000634		
Customer Name	Frauke Seemann			**Customer Unit**	Texas A＆M University		
Lab Sample Collector	Liu Hai			**Lab Sample Receiving Date**	2023-08-09		
Lab Sample Tester	Liu Hai, Yan Pan			**Lab Sample Testing Date**	2023-08-10		
Lab Name	BGI Tech Solutions(Poland) sp. z o.o.			**Lab Address**	Jutrzenki 12 A, 02-230 Warszawa, Poland		
Reported by	Liu Hai	**Inspected by**	Yan Pan	**Approved by**	Yanpan	**Report Date**	2023-08-11

2. Sample Test Method

1.Method of concentration determination: ■Agilent 4150、☐Fragment Analyzer、☐Microplate Reader
2.Method of 28S/18S & 23S/16S test:■Agilent 4150、☐Fragment Analyzer、☐Qsep-400
3.Method of RIN or RQN test:■ Agilent 4150、☐ Fragment Analyzer、☐ Qsep-400

3. Sample Test Result

No.	Sample Name	Sample Number	Tube No.	Concen-tration(ng/μL)	Volume(μL)	Total Mass(μg)	RIN/RQN	28S/18S	Library type	Test result	Remark
1	PFBS4	PL23080900062	1	156	47	7.33	2.7	0.3	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8
2	C4	PL23080900063	1	187	23	4.30	3.5	0.4	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8
3	C5	PL23080900064	1	84.6	23	1.95	2.5	0	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8

4	C6	PL23080900065	1	275	22	6.05	3	0.4	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8
5	PFOS4	PL23080900066	1	271	22	5.96	2.7	0.3	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8
6	PFOS5	PL23080900067	1	204	47	9.59	3	0.4	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8
7	PFOS6	PL23080900068	1	95.7	47	4.50	1.5	0	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8
8	PFHxS4	PL23080900069	1	109	42	4.58	2.6	0.3	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8
9	PFHxS5	PL23080900070	1	124	45	5.58	2.5	0.2	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8
10	PFHxS6	PL23080900071	1	159	46	7.31	4	0.1	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8
11	PFBS5	PL23080900072	1	185	46	8.51	4.7	0.2	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8
12	PFBS6	PL23080900073	1	257	46	11.82	5.3	0.3	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8
13	PFOA4	PL23080900074	1	248	22	5.46	3.7	0.1	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8
14	PFOA5	PL23080900075	1	152	48	7.30	4.5	0.2	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8

15	PFOA6	PL23080900076	1	196	21	4.12	4.2	0.2	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8
16	GenX4	PL23080900077	1	168	23	3.86	2.5	0.1	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8
17	GenX5	PL23080900078	1	122	47	5.73	4.4	0.2	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8
18	GenX6	PL23080900079	1	211	23	4.85	4.5	0.2	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8
19	EE24	PL23080900080	1	312	24	7.49	2.8	0.1	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8
20	EE26	PL23080900081	1	241	23	5.54	4.5	0.2	DNBSEQ stranded mRNA library	Level C	RIN<7,28S/18S<0.8

Note*:

1. The test result notes:

a) Level A means the sample meets the requirements of library construction and sequencing.

b) Level B means the sample does not totally meet the requirements of library construction and sequencing. BGI can try to construct the library but the quality of the sequence is not guaranteed.

c) Level C means the sample does not meet the requirements of library construction and sequencing.

d) Samples with “N/A” results are not evaluated according to sample quality standards due to their specific conditions. Such samples will be executed with customized plans according to their specific situation.

2. The success rate is calculated based on the statistics of samples executed in BGI during recent period of time. Many factors contribute to the success rate of a sample. This data is for reference only, it does not guarantee success rate for individual sample.

3. For Level B or Level C samples, transcriptome library construction has the following risk at least, based on historical statistics in BGI:

a) Low RNA concentration or mass may lead to library construction failure, unsuccessful sequence due to low library production or insufficient sequencing data quantity, affected data randomness and bias introduction.

b) Degradation of sample may lead to a library construction failure, high duplication rate, poor randomness of sequencing data, inaccurate gene expression quantitation.

c) High concentration of RNA may lead to library construction failure caused by inaccurate sampling.

d) A high 5S peak may lead to inaccurate RNA quantification which can lead to poor data quality due to inaccurate sampling.

e) For plant, fungi and bacteria, OD260/280<1.8 and OD260/230<1.8 may indicate presence of impurities that can reduce efficiency of enzyme reaction, affecting the success rate of library construction and quality of sequencing data. These samples have 90% successful rate in library construction.

4.If COs insist on constructing library with Level B or Level C samples, COs shall take the responsibility and risk involved in this matter.

5.The following conditions carry moderate risk, and we can try to construct libraries with them:

a) The RIN or RQN value is slightly under standard, but baseline is smooth.

b) The RIN or RQN value reaches standard, but baseline is rises slightly.

c) The baseline is smooth and RIN value reaches standard, but 5S peak is slightly high.

d) 28S/18S or 23S/16S value is slightly under standard, but the baseline is smooth.

e) Sample generally meets Level A requirements, but total amount is below Level A.

6. Other notes:

a) Meta Strand-Specific Transcriptome library construction is not applicable to complex meta samples such as soil bacterial and ocean microorganism samples.

4. Appendix

Appendix 1: Test results of Agilent 4150 or Fragment Analyzer

Appendix 2: Original information of sample

5. Statement

1. The results shown in this report refer only to the sample of the report unless otherwise stated.
2. This test report cannot be copied partly without the prior written permission of the Lab.

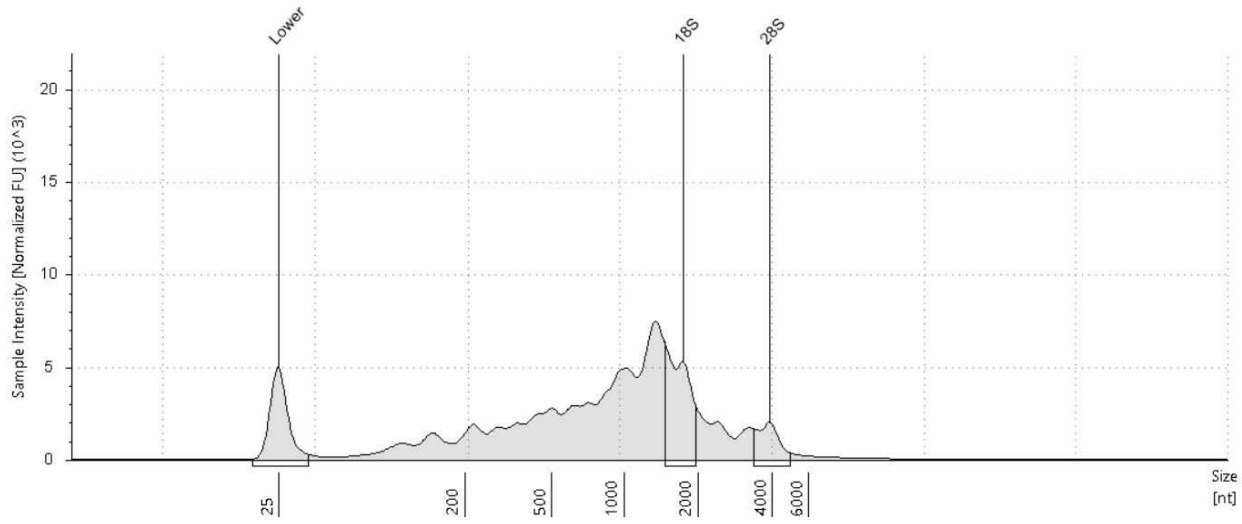
Appendix 1: Test results of Agilent 4150 or Fragment Analyzer or Qsep-400

1.Pre-treatment

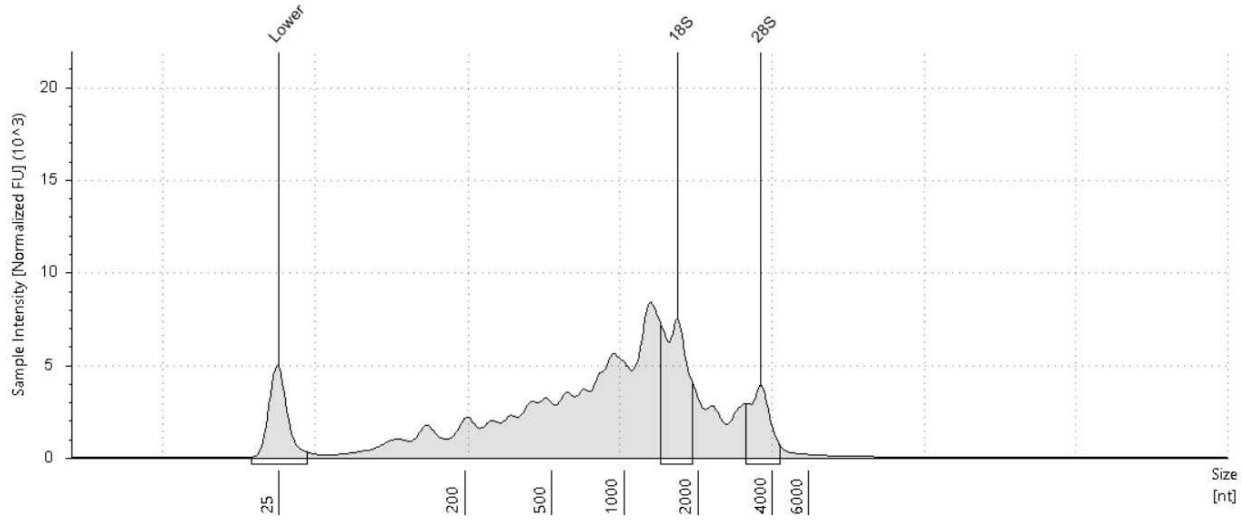
After the sample melted the ice, Add 25 μL of RNase-Free water to part of the sample original tube, centrifuged and fully mixed, take appropriate samples for testing.

2. Test Result

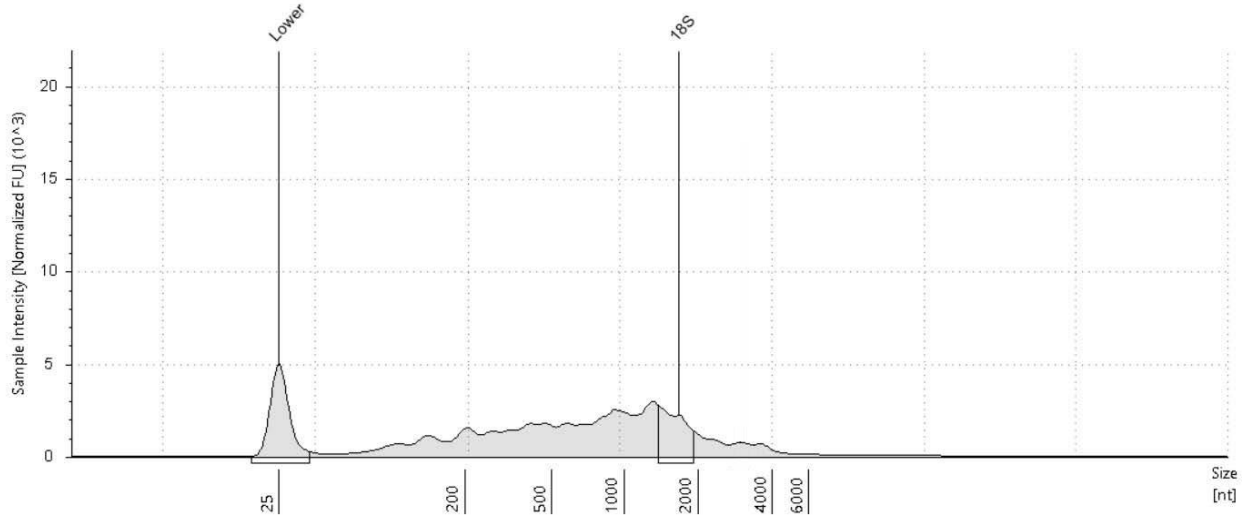
H3: (1)-PFBS4



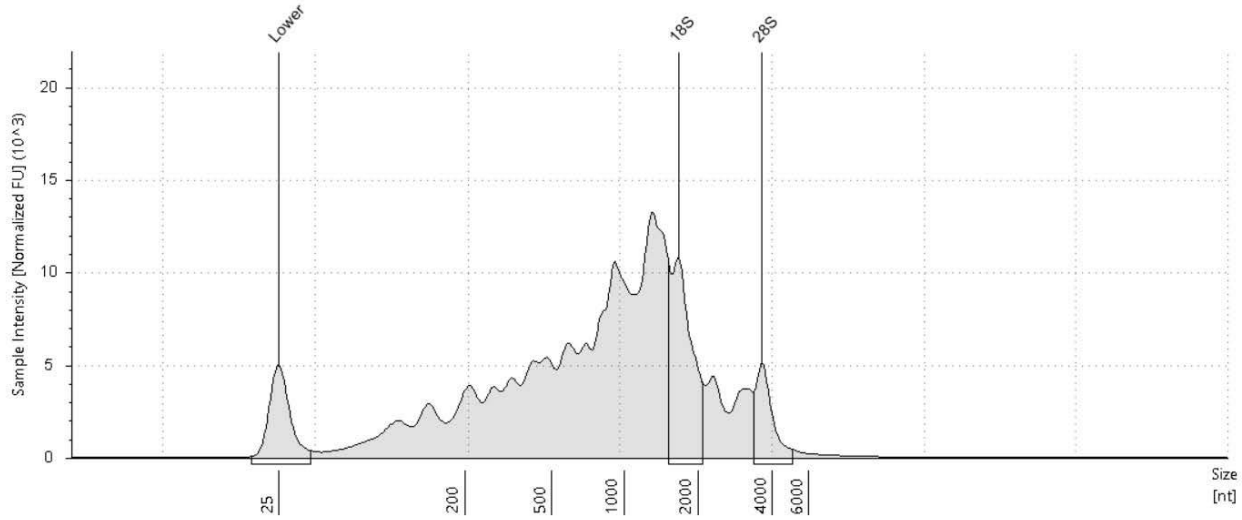
A4: (2)-C4


Lower
18S
28S
Sample Intensity [Normalized FU] (10^3)
20
15
10
5
0
25
200
500
1000
2000
4000
6000
Size
[nt]

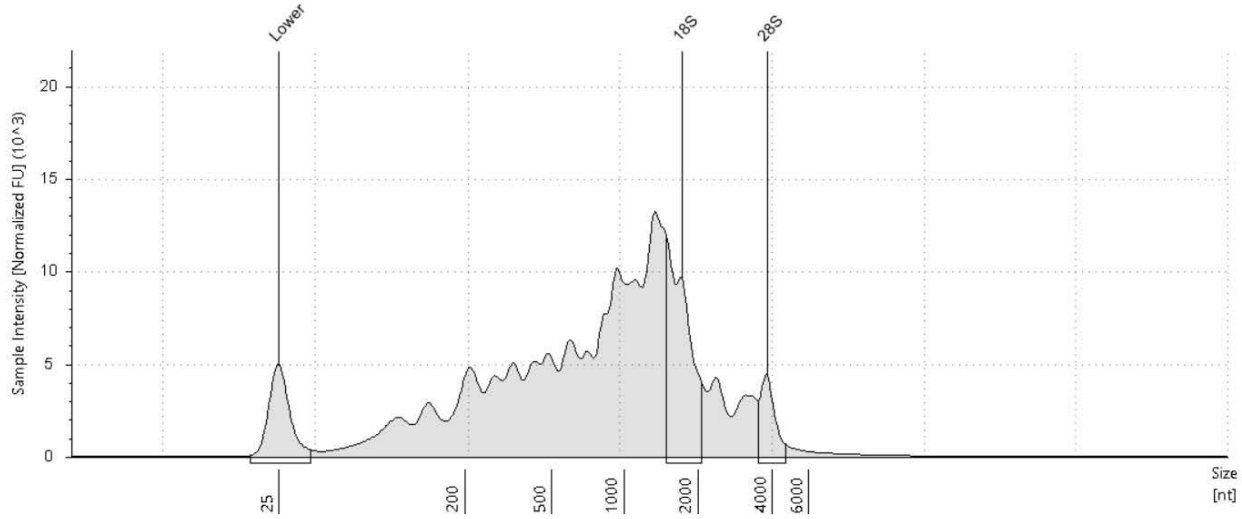
B4: (3)-C5


Lower
18S
Sample Intensity [Normalized FU] (10^3)
20
15
10
5
0
25
200
500
1000
2000
4000
6000
Size
[nt]

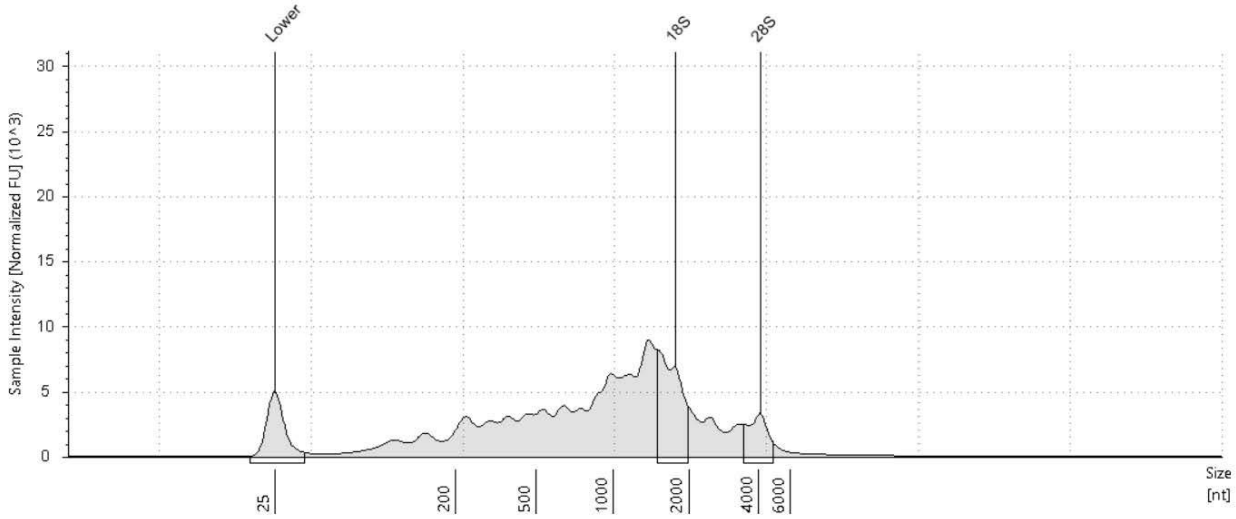
C4: (4)-C6



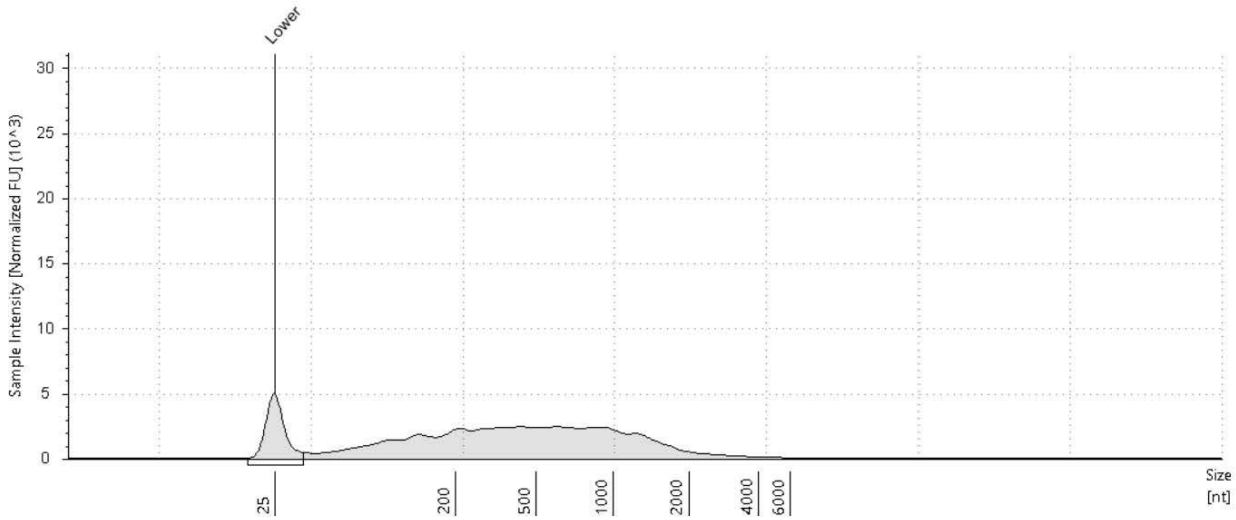
D4: (5)-PFOS4



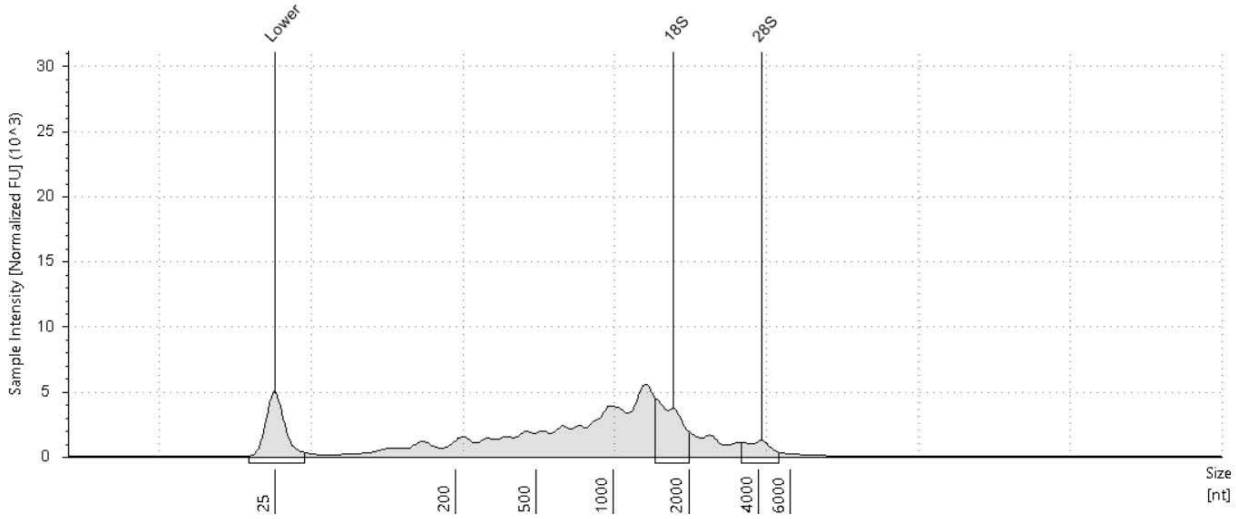
E4: (6)-PFOS5



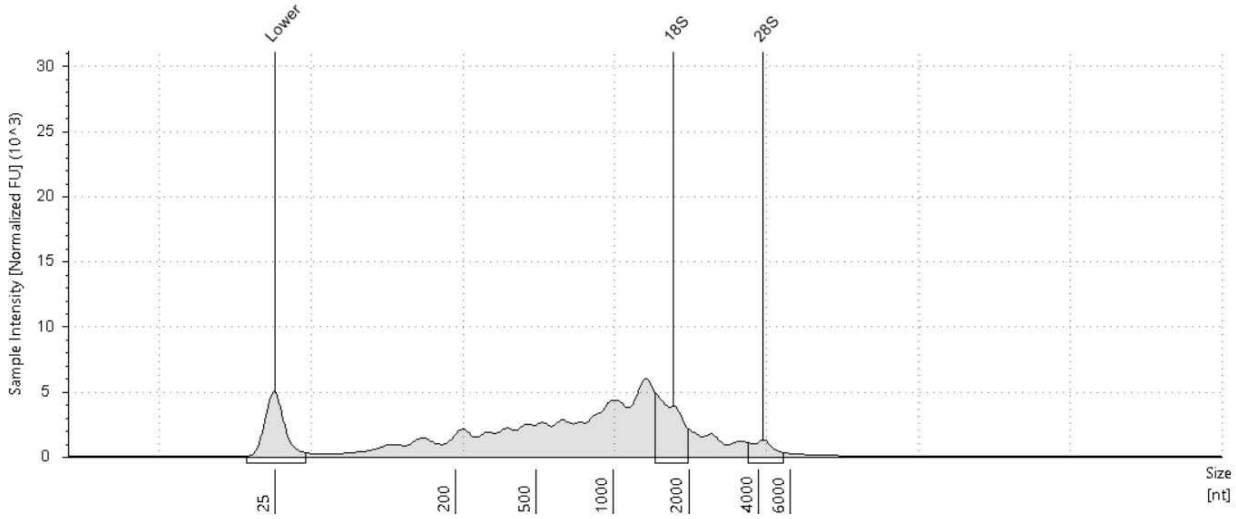
F4: (7)-PFOS6



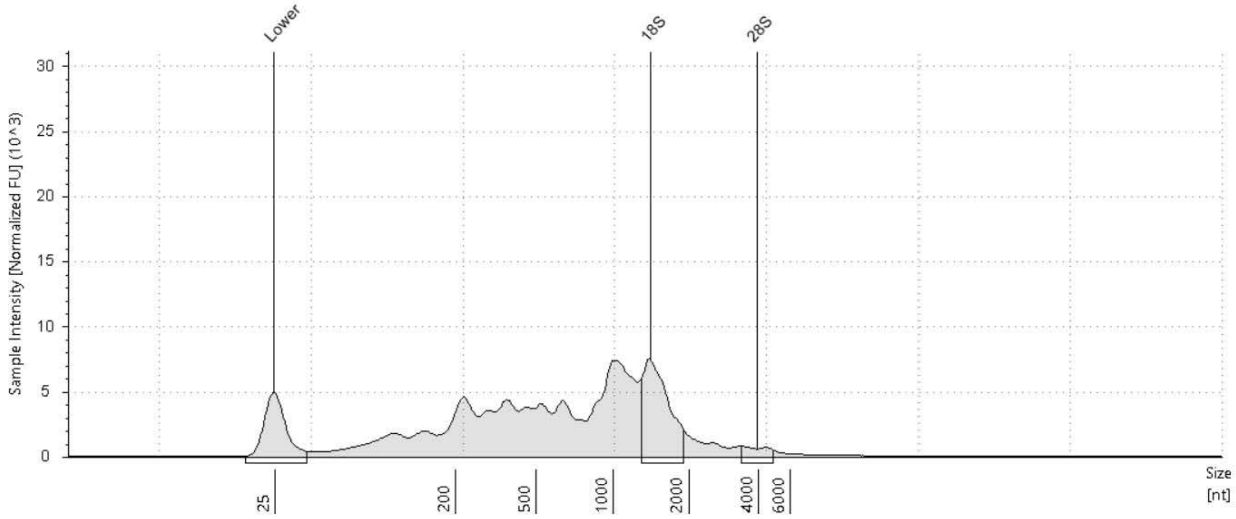
G4: (8)-PFHxS4



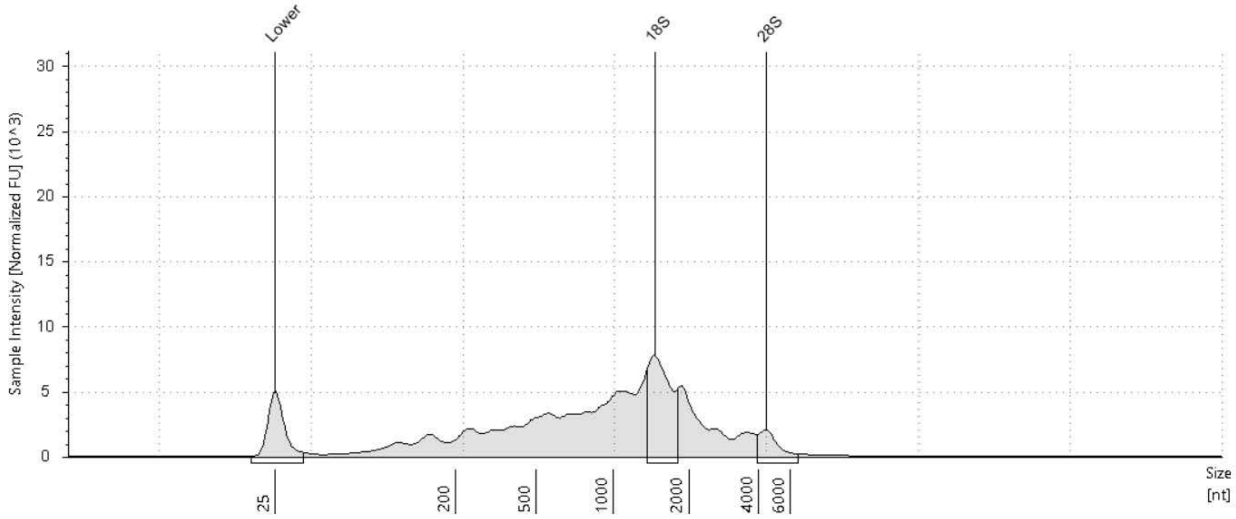
H4: (9)-PFHxS5



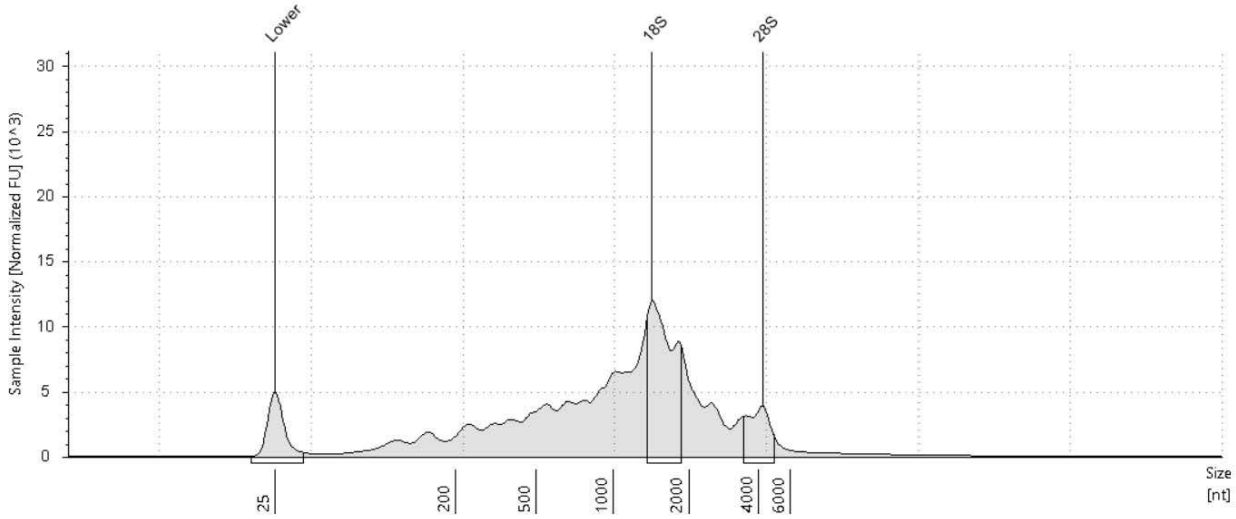
A5: (10)-PFHxS6



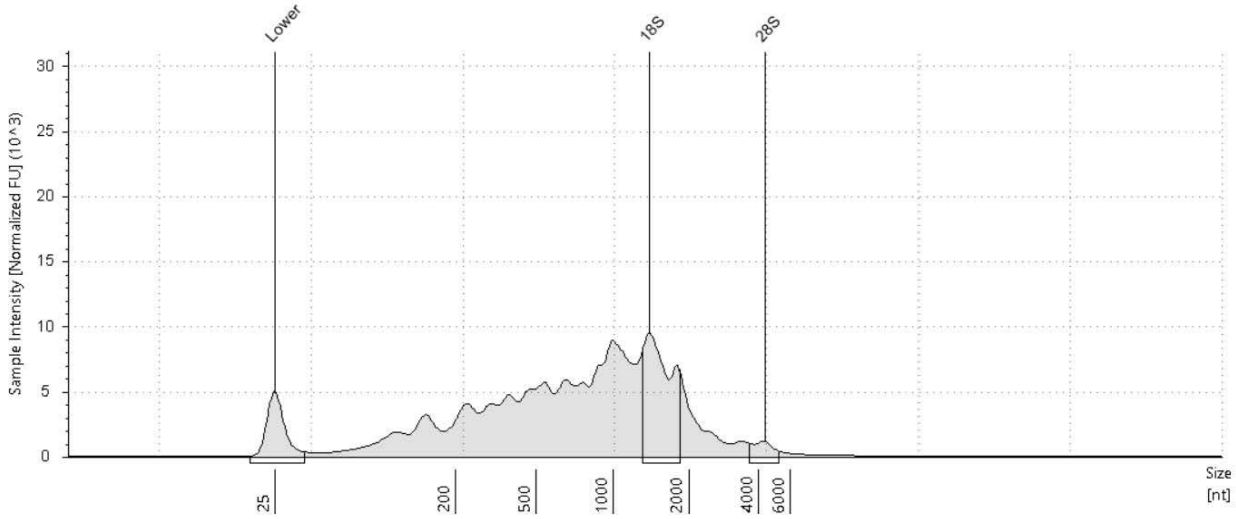
B5: (11)-PFBS5



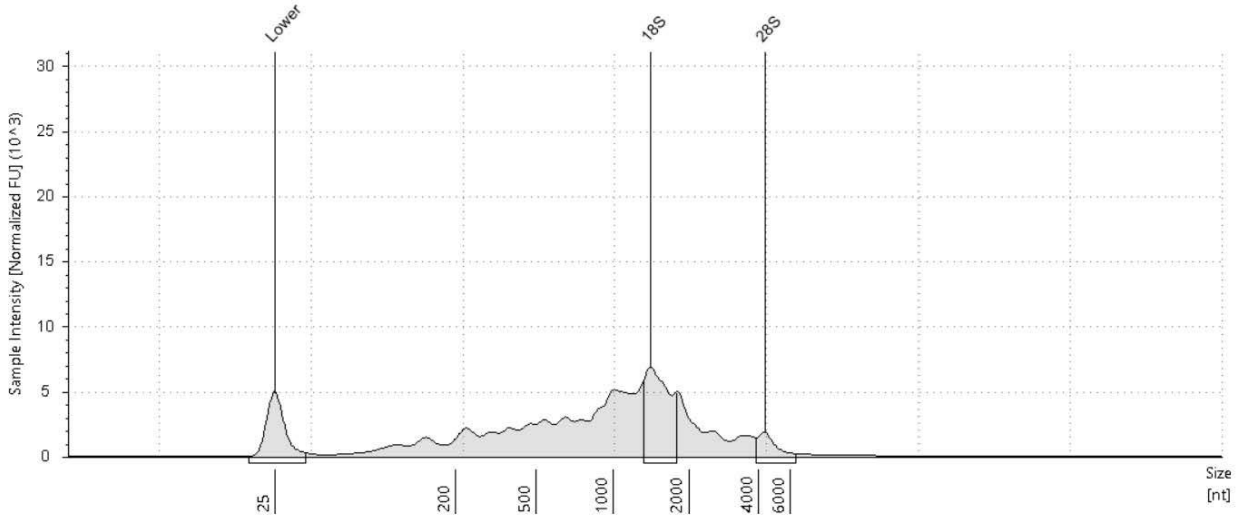
C5: (12)-PFBS6



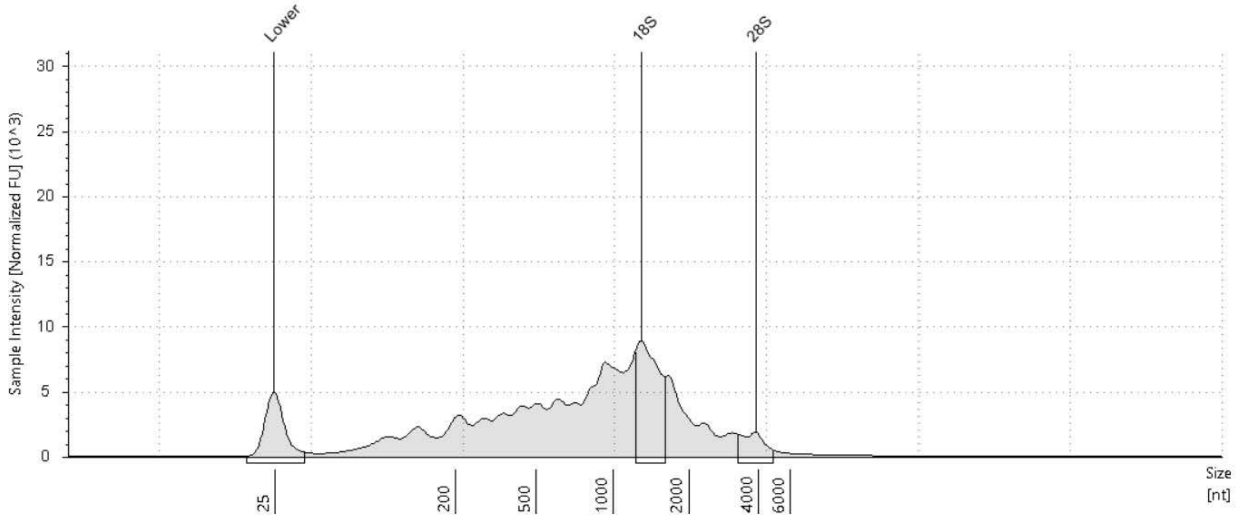
D5: (13)-PFOA4



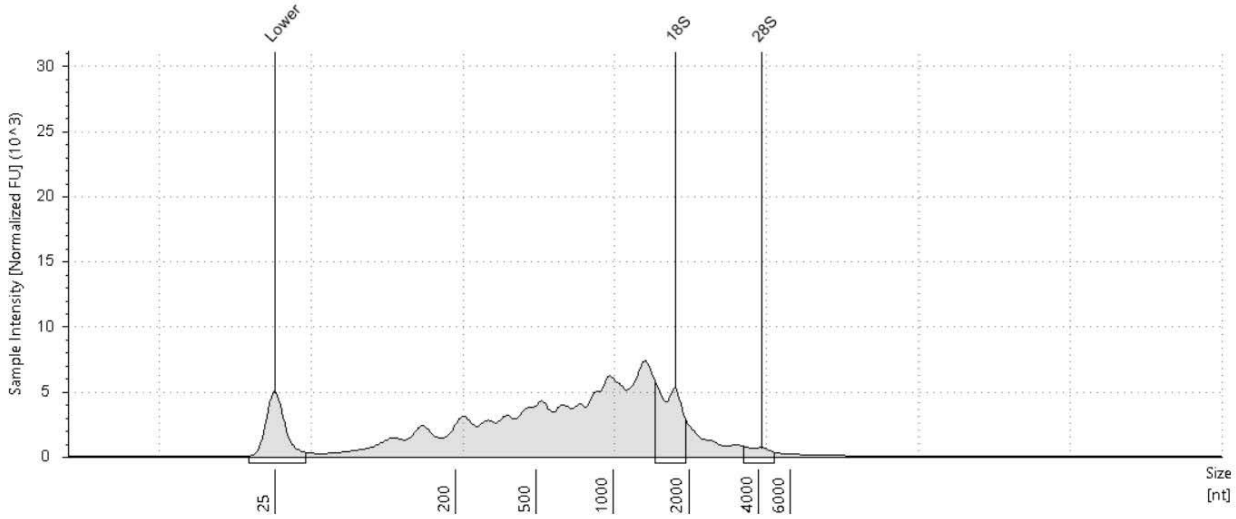
E5: (14)-PFOA5



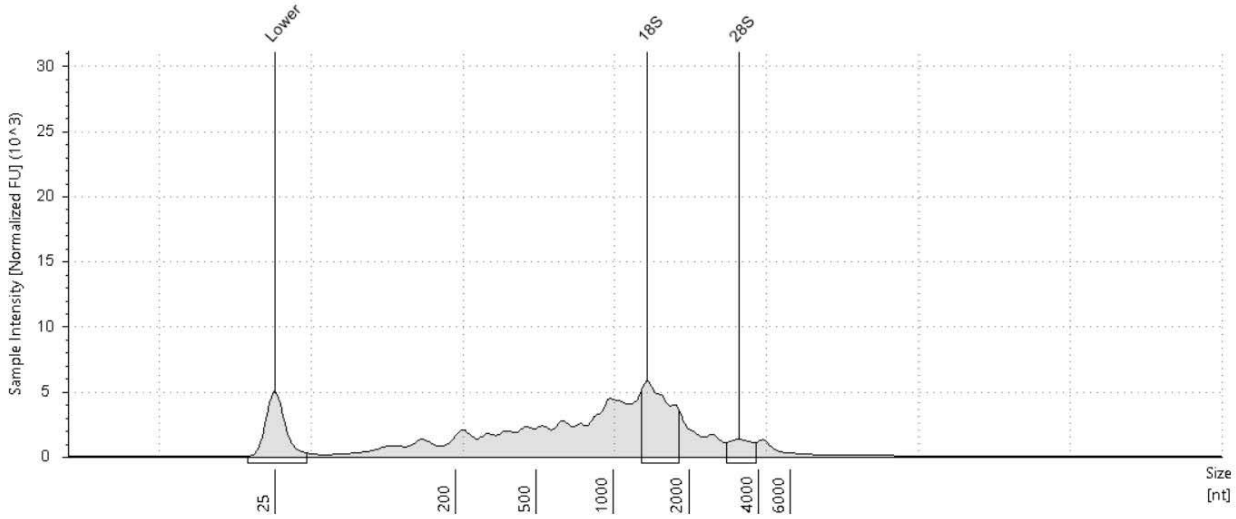
F5: (15)-PFOA6



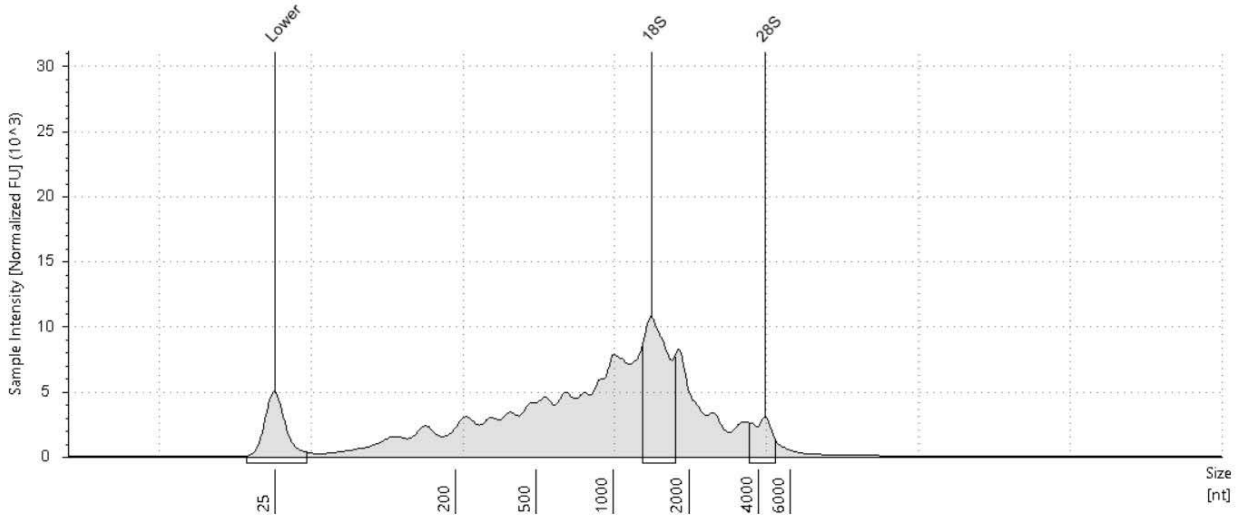
G5: (16)-GenX4



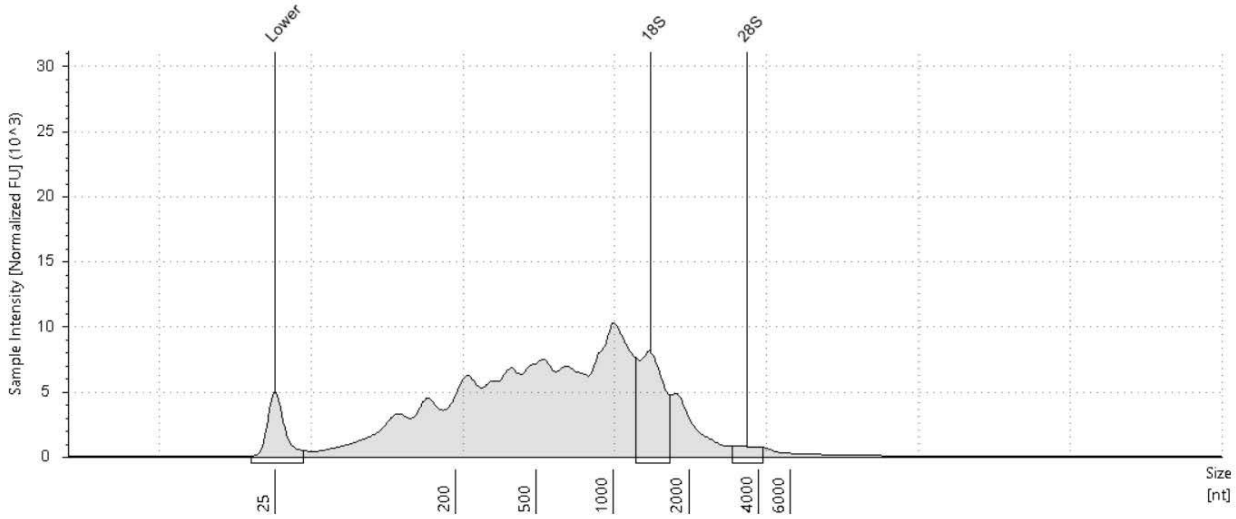
H5: (17)-GenX5



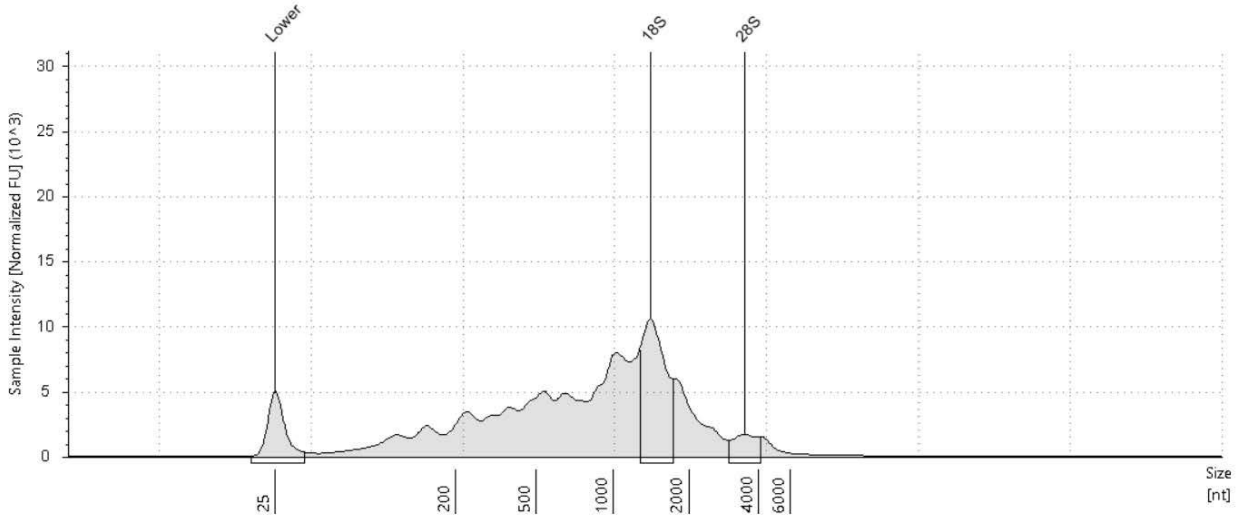
A6: (18)-GenX6



B6: (19)-EE24



C6: (20)-EE26


Lower
18S
28S
Sample Intensity [Normalized FU] (10^3)
30
25
20
15
10
5
0
25
200
500
1000
2000
4000
6000
Size
[nt]

Appendix 2: Original information of sample

Sample Type:									
Total RNA									
Sample status:									
Dissolved in RNase-free Water(or DEPC Treated Water)									
Further Information:									
Sample Name	**Species**	**No. of Tubes**	**Concentration (ng/μL)**	**Volume (μL)**	**Total Quantity (μg)**	**OD260/280**	**OD260/230**	**28S/18S**	**Remark**
C4	Oryzias melastigma	1	291.7	25	7.293	1.98	2.18		
C5	Oryzias melastigma	1	279.1	25	6.978	1.94	1.21		
C6	Oryzias melastigma	1	437.5	25	10.938	1.98	2.26		
PFOS4	Oryzias melastigma	1	479.2	25	11.980	1.97	2.26		
PFOS5	Oryzias melastigma	1	663.8	25	16.595	1.98	2.13		
PFOS6	Oryzias melastigma	1	596.5	25	14.913	1.97	2.24		
PFHxS4	Oryzias melastigma	1	554.1	25	13.853	1.99	2.39		
PFHxS5	Oryzias melastigma	1	564.4	25	14.110	1.98	2.2		
PFHxS6	Oryzias melastigma	1	793.1	25	19.828	1.99	1.64		
PFBS4	Oryzias melastigma	1	584.3	25	14.608	1.97	2.22		
PFBS5	Oryzias melastigma	1	583.9	25	14.598	1.97	2.13		

PFBS6	Oryzias melastigma	1	748.6	25	18.715	1.99	1.8		
PFOA4	Oryzias melastigma	1	327.7	25	8.193	1.92	1.19		
PFOA5	Oryzias melastigma	1	541.6	25	13.540	1.98	2.1		
PFOA6	Oryzias melastigma	1	459.5	25	11.488	1.97	2.02		
GenX4	Oryzias melastigma	1	308.3	25	7.708	1.92	2.14		
GenX5	Oryzias melastigma	1	585	25	14.625	1.97	2.03		
GenX6	Oryzias melastigma	1	492	25	12.300	1.86	1.43		
EE24	Oryzias melastigma	1	415.7	25	10.393	1.96	1.6		
EE26	Oryzias melastigma	1	464.3	25	11.608	1.97	1.83		

----------Report End----------